Edge-Cell Graph Encoding and Fixed-Point Arithmetic for FPGA Growing Neural Gas

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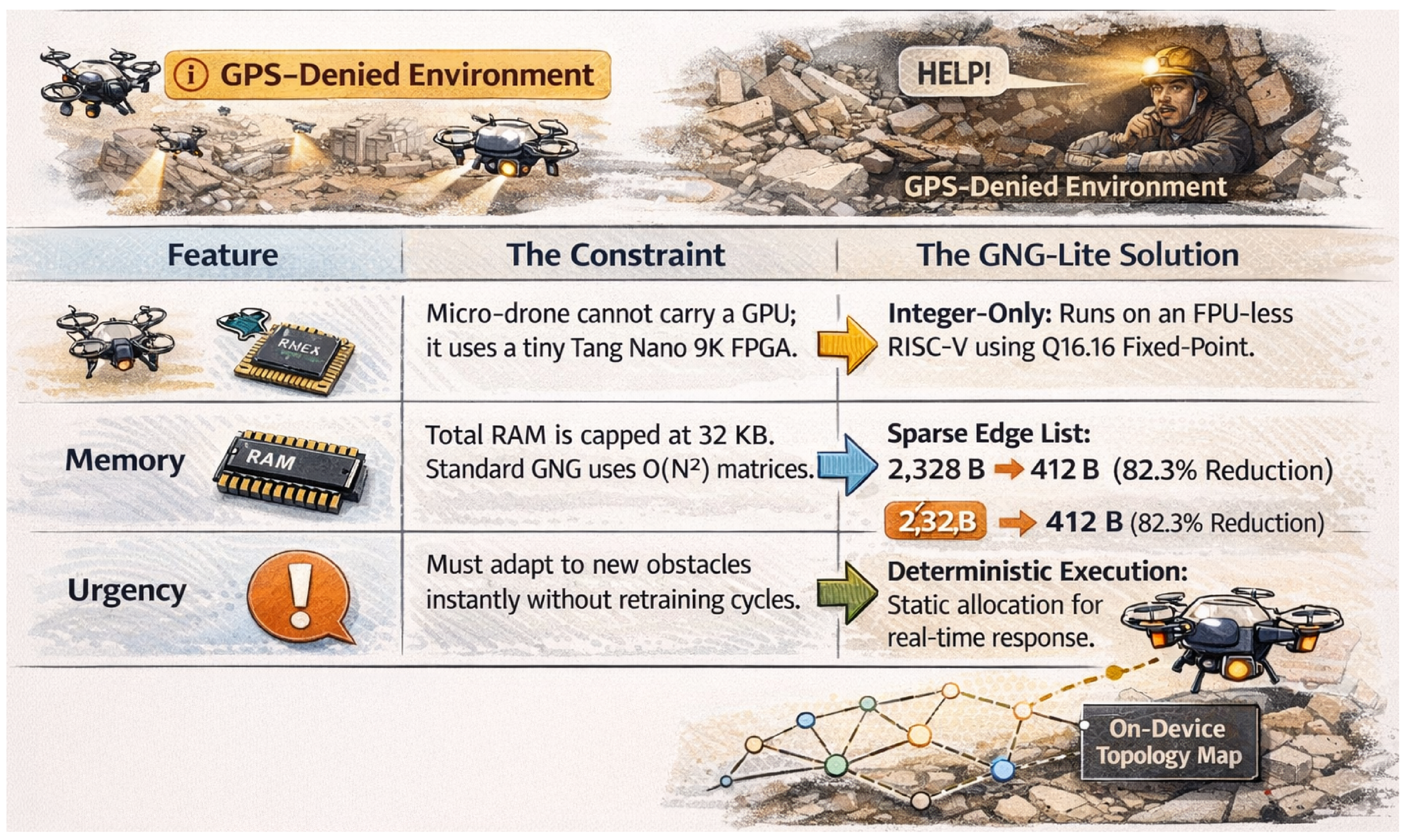
Motivation & Problem

Edge robots, micro-UAV swarms, and field devices need **on-board unsupervised adaptation** (topology formation, novelty detection) but cannot stream raw data to the cloud. Learning must run **locally** with **bounded memory** and **deterministic per-sample timing**.

Growing Neural Gas (GNG) is ideal — it learns data topology incrementally, sample-by-sample, with no labels or retraining. But classic GNG clashes with small-FPGA limits:

- **Dense adjacency matrix** scales $O(N^2)$: a 40×40 8-bit matrix needs **1,600 B** — it dominates BRAM.
- **Floating-point** arithmetic needs FPU blocks or slow iterative units.



Our Contributions

A fully integer, single-FSM GNG engine in synthesizable VHDL on the **Sipeed Tang Nano 9K (Gowin GW1NR-9C)** — **no soft-core CPU, no FPU**.

1. **Edge-cell encoding**: upper-triangular half-adjacency as 8-bit *age+1* values (0 = no edge). **780 B** vs. 1,600 B.
2. **Shift-based Q8/Q16** learning rates ⇒ single-cycle DSP multiplies, no FPU.
3. **Fused update pass**: error decay + edge aging + neighbor move + pruning in *one* sequential scan.
4. **Degree counters** for $O(N_{\max})$ isolated-node detection.

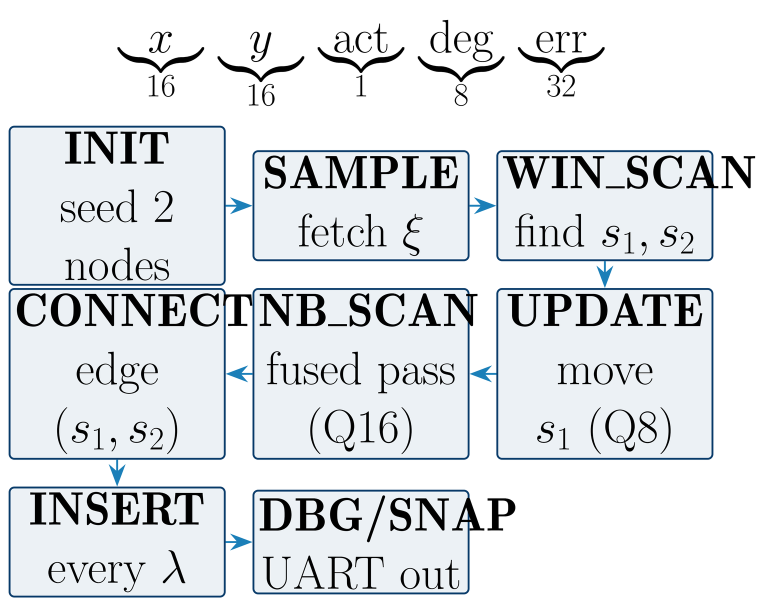
GNG Algorithm Core

Maintain a graph $G = (V, E)$ with weight vectors $w_i \in \mathbb{R}^2$. For each input sample ξ:

1. Find winner $s_1 = \arg\min_i \|\xi - w_i\|^2$ and runner-up s_2.
2. Accumulate / decay winner error; move s_1: $w_{s_1} \leftarrow w_{s_1} + \epsilon_w(\xi - w_{s_1})$.
3. Age edges of s_1; move neighbours; **prune over-age** edges.
4. Connect (s_1, s_2), reset age $= 0$.
5. Every λ steps insert node r at the highest-error region; split the error.

FPGA Architecture

Single synchronous **FSM datapath** at **27 MHz** (crystal, no PLL). Three BRAMs: node, edge, dataset. **80-bit node word**:



Each BRAM read-modify-write takes ≥ 3 cycles (address, wait, evaluate).

Key Idea: Edge-Cell Encoding

Store only the **upper triangle** of the undirected adjacency, one byte per potential edge, with an **age+1** code:

$$E_{\text{cells}} = \frac{N_{\max}(N_{\max}-1)}{2} = \frac{40 \times 39}{2} = \textbf{780 bytes}$$

- `cell`$= 0$: no edge • `cell`$= v > 0$: active, true age $= v - 1$.
- No separate *active* bit; over-age prune = one compare (`v > 51`).
- Linear index $\texttt{idx}(i,j) = \frac{i(2N_{\max}-i-1)}{2} + (j - i - 1)$, $i < j$.

Fused NB_SCAN Pass

One sequential scan over $N_{\max}$ slots folds together: **global error decay** + **edge aging** + **neighbor movement** + **over-age pruning**, replacing a separate full-array decay sweep.

Integer Fixed-Point Arithmetic

16-bit signed coordinates in $[-1000, 1000]$ (min-max norm $\times 1000$).
Distance maps onto 9×9 DSPs, no overflow:

$$d^2 = \Delta x^2 + \Delta y^2 \quad \text{(35-bit unsigned)}$$

Winner (Q8) and **neighbor (Q16)** moves use shift-scaled rates:

$$\Delta x_{s_1} = \left\lfloor \frac{77\,(\xi_x - x_{s_1})}{2^8} \right\rfloor \ (\epsilon_w \approx 0.301), \qquad \Delta x_n = \left\lfloor \frac{66\,(\xi_x - x_n)}{2^{16}} \right\rfloor \ (\epsilon_n \approx 0.001)$$

Error decay is a single right-shift: $\text{err}_n \mathrel{-}= \text{err}_n \gg 8$ ($\beta \approx 255/256$) — no multiplier needed.

Results

Memory footprint ($N_{\max} = 40$)

	Edge	Total
Dense adj.	1600 B	2000 B
Edge-cell	**780 B**	**1180 B**
Reduction	**51%**	**41%**

FPGA utilization (GW1NR-9C)

Resource	Used	%
LUT	3205	38%
FF	1408	22%
BSRAM	6	24%
DSP	5	50%

Topology quality & runtime (27 MHz)

Dataset	Nodes	Edges	QE	TE	μs/sample
Two Moons	40	37	0.146	0.00	624
Concentric Circles	40	40	0.200	0.04	627

Deterministic ~625 μs/sample ⇒ ~1,600 samples/s.

Learned Topologies

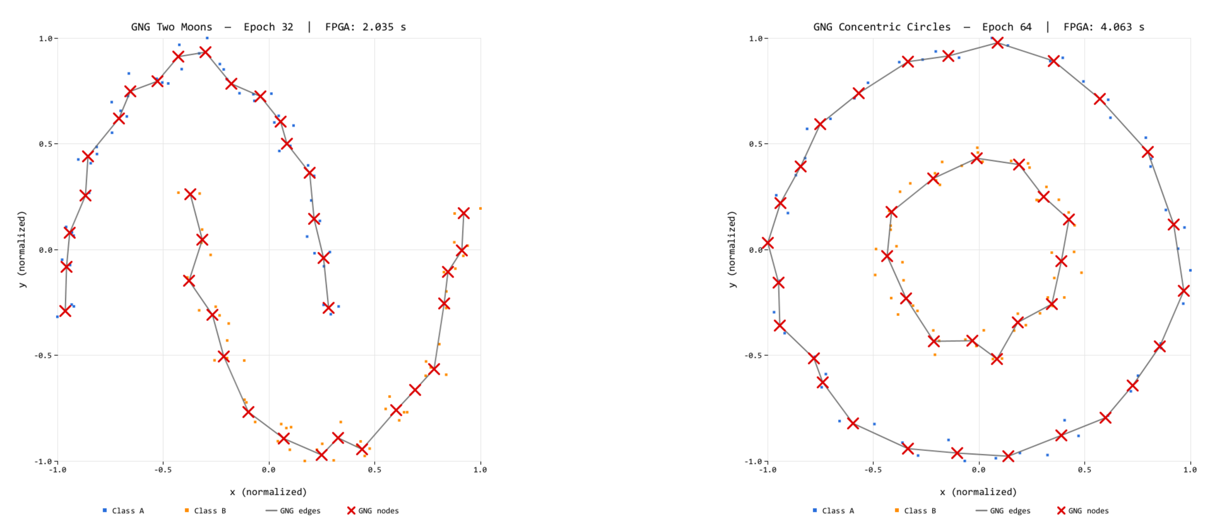


Fig. 2: Two Moons (left) and Concentric Circles (right): learned GNG graph, $N_{\max} = 40$.

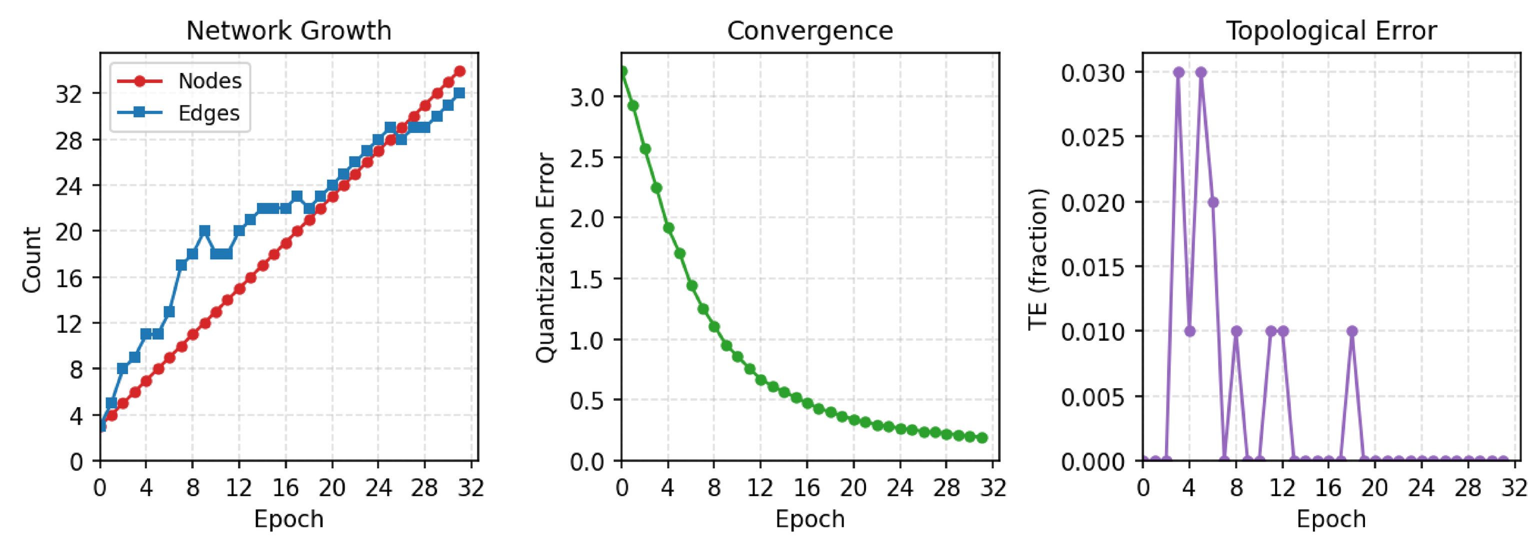


Fig. 3: Training dynamics: node/edge growth, QE convergence, TE vs. epoch.